Knee MRI Dictation Cheat Sheet - Expanded Findings & Impressions

Copy-ready language. Replace bracketed text. Use as a workstation reference, not a substitute for surgical correlation or local report style.

Core Measurements

System	Use
Modified Outerbridge / ICRS	Cartilage grading: 0 normal, 1 softening, 2 <50% depth, 3 >50% depth, 4 full-thickness loss.
Meniscal extrusion	>=3 mm beyond tibial margin is significant; associated with root tears and accelerated OA.
ACL tunnel sizing	Femoral tunnel [X] mm, tibial tunnel [X] mm; report widening if >1 mm increase from baseline.
TT-TG distance	Normal <15 mm, borderline 15-20 mm, abnormal >20 mm; relevant for patellar instability.
Insall-Salvati ratio	Patellar tendon length / patellar diagonal: normal 0.8-1.2; >1.2 patella alta, <0.8 patella baja.
Caton-Deschamps index	Distance from inferior patellar articular surface to anterosuperior tibial plateau / patellar articular surface length: normal 0.6-1.2.

How to Find It — Landmark-Based Identification

This section teaches you to *locate* each knee structure by its anatomic landmark and the plane/sequence that displays it best, what a normal version looks like, and the classic normal variants and mimics that fool trainees. Build the habit of confirming every finding on at least two planes before you call it.

The Three Planes — What Each Is Best For

💡 PEARL **Match the plane to the structure**

Each plane has a job. Read every structure on its best plane first, then confirm on a second plane.

- **Sagittal**: menisci (body as "bowtie", horns in profile), **ACL** and **PCL**, patellar/quad tendons, trochlear cartilage, Hoffa fat pad.
- **Coronal**: meniscal body and roots, **MCL** (superficial + deep), **LCL**, meniscal extrusion, tibiofemoral cartilage, collateral origins.
- **Axial**: **patellofemoral** joint (tracking, trochlear shape, cartilage), **MPFL** and retinacula, popliteus/PLC relationships, cysts.

💡 PEARL **Sequence logic**

Fluid-sensitive fat-suppressed sequences (**PD/T2 FS**) show edema, tears, and cartilage defects; non-fat-suppressed **PD/T1** show anatomy and fibrocartilage morphology with better signal-to-noise.

- Meniscal signal grading is best judged on short-TE images (**PD**), where the meniscus is uniformly dark.

Menisci — Body, Horns, Roots

🔍 HOW TO FIND IT **Meniscal body — the "bowtie"**

On **sagittal** peripheral slices, the meniscal body appears as a **bowtie**: a continuous dark band connecting anterior and posterior horns.

- Normal: count **≥2 consecutive bowties** for the body (slice thickness/spacing dependent, but two is the working rule).
- Move centrally and the bowtie "opens" into separate triangular anterior and posterior horns.

💡 PEARL **Counting bowties for a bucket-handle tear**

The lateral and medial menisci should each show ~**2 bowtie slices**. **Fewer than 2 bowties** suggests a displaced bucket-handle fragment (meniscus is shortened in the AP/peripheral dimension).

- Corroborate with the **double-PCL sign**, the **double anterior horn sign**, and a **fragment in the intercondylar notch**.

🔍 HOW TO FIND IT **Posterior horns and the root attachments**

On **sagittal**, the **posterior horn of the medial meniscus** is normally the **largest** meniscal segment; the lateral posterior horn is smaller and more symmetric with its anterior horn.

- **Roots** attach to the **central tibial plateau** near the tibial eminence; find them on **coronal** and central **sagittal/axial**.
- Normal root: dark fibers inserting at the plateau without fluid signal undercutting them.

⚠️ PITFALL **Root tear and the "ghost meniscus"**

A **medial posterior root tear** shows a radial/vertical cleft at the insertion, with the body **extruding** medially; on a central slice the posterior horn may seem to vanish (**"ghost meniscus"**).

- Always check the **coronal** for extrusion when the posterior root looks abnormal.

📏 MEASURE **Meniscal extrusion**

On the **coronal** slice through the mid-joint (at the medial collateral level / tibial eminence), measure the meniscus from the edge of the tibial plateau (exclude osteophytes).

- **≥3 mm** of extrusion is considered significant ("major" extrusion) and is associated with root tears and degeneration.

📏 MEASURE **Rules for calling a meniscal tear**

Call a tear when **signal unequivocally reaches an articular surface**.

- Require surface-reaching signal on **≥2 consecutive slices** (the "two-slice touch / two-touch rule") to call a tear with confidence.
- **Grade 1** (intrasubstance globular) and **grade 2** (linear, not reaching surface) signal are **degeneration, not tears**.
- Note morphology: horizontal, vertical-longitudinal, radial, complex, displaced/bucket-handle.

Normal Meniscal Structures That Mimic Tears

⚠️ PITFALL **Meniscofemoral ligaments — Humphrey and Wrisberg**

These run from the **lateral meniscus posterior horn** to the **lateral aspect of the medial femoral condyle**, flanking the PCL.

- **Humphrey** = **anterior** to PCL; **Wrisberg** = **posterior** to PCL (mnemonic: "**H** before **W**" = anterior before posterior).
- Their origin creates a **cleft of fluid signal** at the lateral posterior horn — do **not** call this a tear.

⚠️ PITFALL **Popliteomeniscal fascicles and the popliteus hiatus**

The **popliteus tendon** passes through the **hiatus** at the **lateral meniscus posterior horn**, tethered by superior and inferior **popliteomeniscal fascicles**.

- The hiatus produces normal fluid signal/cleft at the **posterolateral** meniscus — a classic **pseudotear**.
- Orientation is **oblique** and follows the tendon; a true tear is typically vertical or horizontal and not aligned with the hiatus.

⚠️ PITFALL **Transverse (intermeniscal) ligament**

Connects the **anterior horns** of the medial and lateral menisci across the anterior knee.

- On **sagittal**, its junction with the anterior horn mimics an **anterior horn tear** ("pseudotear"); trace it on **axial** to confirm it is a ligament coursing through Hoffa fat.

⚠️ PITFALL **Meniscal flounce**

A **smooth, wavy fold** of the (usually medial) meniscal free edge seen on a single sagittal image.

- It is a **normal**, often position-dependent finding — **not** a tear. Confirm smooth contour and absence of surface-reaching signal.

Cruciate Ligaments

🔍 HOW TO FIND IT **Anterior cruciate ligament**

On **sagittal**, find the ligament paralleling the **roof of the intercondylar notch (Blumensaat line)**; confirm fibers on coronal/axial.

- Normal: continuous, taut, **parallel to Blumensaat line**; fibers may appear slightly striated (especially the two bundles) but should be continuous and not wavy.
- Origin: posteromedial **lateral femoral condyle**; insertion: **anterior tibial spine** region.

💡 PEARL **ACL tear — primary and secondary signs**

Primary signs are intrinsic; secondary signs are downstream geometry changes.

- **Primary**: discontinuity/non-visualization, abnormal **T2 signal** within the ligament, **wavy or horizontal** fibers (angle below Blumensaat), abnormal **flat** ACL slope.
- **Secondary**: **pivot-shift bone bruises** (lateral femoral condyle + posterolateral tibial plateau), **anterior tibial translation** (>~7 mm), **deepened lateral femoral sulcus**, **uncovered/posteriorly displaced posterior horn of lateral meniscus**, **PCL buckling**, **Segond fracture**.

🔍 HOW TO FIND IT **Posterior cruciate ligament**

On **sagittal**, the PCL is a **thick, uniformly dark, arcuate (curved) band** from the **medial femoral condyle** to the **posterior tibia** below the joint line.

- Normal: smooth curve; can become **more curved/buckled** when the ACL is torn (a secondary ACL sign), so judge PCL signal, not just shape.

Medial and Lateral Collateral Ligaments

🔍 HOW TO FIND IT **Medial (tibial) collateral ligament**

Best on **coronal**. The **superficial MCL** is a long, thin, dark band from the **medial femoral epicondyle** to the **tibia ~5–7 cm below the joint line**.

- The **deep MCL** (meniscofemoral + meniscotibial portions) attaches to the **medial meniscus body**; normal fat/bursa separates it from the superficial layer — do not mistake that fat plane for a tear.
- Grade sprains: **I** = perifascicular edema, intact fibers; **II** = partial thickness; **III** = full-thickness disruption.

🔍 HOW TO FIND IT **Lateral (fibular) collateral ligament**

On **coronal** (often a slightly posterior slice), trace the cord-like **LCL** from the **lateral femoral epicondyle** to the **fibular head**, where it joins the biceps femoris.

- It is **extra-articular** and separate from the lateral meniscus (no meniscal attachment — unlike the MCL).

Posterolateral Corner

📍 LANDMARK **The fibular head and styloid — PLC hub**

The **fibular head/styloid** anchors most PLC structures; orient yourself here on **coronal** and **axial**.

- **LCL** → lateral fibular head.
- **Popliteofibular ligament** → from popliteus musculotendinous junction to the **fibular styloid** (best on coronal oblique/axial).
- **Biceps femoris–IT band conjoined insertion** → posterolateral fibular head / Gerdy tubercle region.

🔍 HOW TO FIND IT **Popliteus tendon and hiatus**

Find the **popliteus tendon** in its groove on the **lateral femoral condyle** (coronal/axial), passing intra-articularly through the **popliteus hiatus** at the posterolateral meniscus toward the muscle belly.

- Normal: smooth dark tendon in its sulcus; the surrounding hiatal fluid is normal (see pseudotear pitfall above).

🔍 HOW TO FIND IT **Arcuate and fabellofibular ligaments**

These thin posterolateral structures attach to the **fibular styloid**.

- The **arcuate sign** = avulsion fracture of the **fibular styloid**, a marker of PLC injury — look for a small bony fragment on coronal/axial and scrutinize the rest of the PLC.

Extensor Mechanism and Patellofemoral Joint

🔍 HOW TO FIND IT **Quadriceps and patellar tendons**

Both are best profiled on **sagittal**.

- **Quadriceps tendon**: multilaminar striated band inserting on the **superior patella** (the lamination is normal).
- **Patellar tendon**: uniformly dark band from the **inferior patellar pole** to the **tibial tubercle**; normal thickness up to ~**7 mm**.

📏 MEASURE **Patella alta/baja — Insall-Salvati and Caton-Deschamps**

Assess on a **sagittal** image.

- **Insall-Salvati** = **patellar tendon length ÷ patellar (bone) length**. Normal ~**0.8–1.2**; **>1.2 = alta**, **<0.8 = baja**.
- **Caton-Deschamps** = distance from inferior articular cartilage margin of patella to anterosuperior tibia, divided by patellar articular surface length. Normal ~**0.6–1.2**; **>1.2 = alta**.

🔍 HOW TO FIND IT **Trochlea and trochlear dysplasia (Dejour features)**

Evaluate the **trochlear groove** on **axial** slices a few mm below the roof, where the groove is normally a clean V.

- **Dejour features** of dysplasia: **crossing sign**, **trochlear bump/spur (supratrochlear)**, **double-contour sign**, and a **shallow/flat or convex groove**.

📏 MEASURE **Trochlear sulcus angle and depth**

On **axial** at the proximal trochlea:

- **Sulcus angle** normally **≤145°**; a wider (flatter) angle indicates dysplasia.
- Trochlear depth **<3 mm** suggests dysplasia.

📏 MEASURE **TT-TG distance (tibial tubercle–trochlear groove)**

Measure on **axial** by superimposing two slices: one through the deepest point of the **trochlear groove** and one through the **tibial tubercle** at the patellar tendon insertion. Measure the **mediolateral distance** between the two points, perpendicular to a line along the posterior femoral condyles.

- Normal **<15 mm**; **≥15–20 mm** is abnormal and **>20 mm** strongly favors lateralized tracking/instability.

🔍 HOW TO FIND IT **Medial patellofemoral ligament (MPFL)**

On **axial**, trace the **MPFL** from its femoral origin between the **adductor tubercle and medial epicondyle (Schöttle point)** to the **superomedial patella** (proximal ~⅓ of the medial patellar border).

- After **lateral patellar dislocation**, look for MPFL disruption at the femoral origin plus the classic **bone-bruise pair**: **medial patella** and **anterolateral/inferior lateral femoral condyle**, often with an osteochondral fragment.

Cartilage — Systematic Compartment Mapping

💡 PEARL **Cover every cartilage surface in order**

Map cartilage by compartment so nothing is missed; use fluid-sensitive FS sequences and grade defects by depth (fibrillation → partial → full-thickness → subchondral involvement).

- **Patellofemoral**: patellar facets (medial, lateral, odd) on **axial**; trochlea on **axial + sagittal**.
- **Medial compartment**: medial femoral condyle + medial tibial plateau on **coronal + sagittal**.
- **Lateral compartment**: lateral femoral condyle + lateral tibial plateau on **coronal + sagittal**.

⚠️ PITFALL **Acute vs chronic osteochondral defect — say which**

Date the lesion from the bone, not just the cartilage. The wording changes management.

- **Acute** (osteochondral fracture/injury): **subchondral marrow edema**, a sharp **non-sclerotic** margin, ± a subchondral **fracture line**, and often an effusion with a **lipohemarthrosis** (fat–fluid level = intra-articular fracture). Hunt for a **displaced fragment / loose body**.
- **Chronic** (e.g., old OCD/defect): **sclerotic rim**, **subchondral cysts**, and little or no marrow edema.

📏 MEASURE **Is the osteochondral fragment unstable?**

Call instability when you see any of: **fluid signal fully undermining the fragment**, a **displaced fragment** or a **loose body**, a **focal cartilage breach with fluid entering**, or (chronic) **subchondral cysts >5 mm**. Report size as **AP × transverse × depth (mm)** and state stability explicitly — it drives surgical decision-making.

Classic Landmarks and Associations

📍 LANDMARK **Segond fracture**

A small **vertical avulsion fragment off the lateral tibial plateau** (lateral capsular/anterolateral ligament region), best seen on **coronal**.

- **Highly associated with ACL tears** (and frequently meniscal injury) — finding it should send you straight to the ACL.

🔍 HOW TO FIND IT **Baker (popliteal) cyst**

Look on **axial** posteromedially: a Baker cyst has a **neck between the semimembranosus tendon and the medial head of gastrocnemius**.

- That specific neck location distinguishes it from other posterior cystic lesions; rupture tracks fluid distally into the calf.

📍 LANDMARK **Pivot-shift bone bruise pattern**

The classic **ACL-injury contusion pair**: **lateral femoral condyle (sulcus terminalis)** and **posterolateral tibial plateau**.

- Recognizing this pattern should prompt careful ACL, lateral meniscus, and PLC review.

🔍 HOW TO FIND IT **Synovial plicae**

Plicae are normal synovial folds; name them by location.

- **Mediopatellar (medial) plica**: along the medial joint, best on **axial** — can become symptomatic/thickened and abrade the medial femoral condyle cartilage.
- **Suprapatellar** and **infrapatellar (ligamentum mucosum)** plicae: identify on **sagittal**; the infrapatellar plica runs through Hoffa fat toward the notch and should not be mistaken for the ACL or a loose body.

Modified Outerbridge Chondromalacia Grading - Copy-Ready

Use this for any articular surface. Replace `[surface/compartment]` with the specific location: patellar median ridge, medial femoral condyle weightbearing surface, anterosuperior acetabulum, radiocapitellar joint, tibiotalar dome, first MTP joint, etc.

Grade	MRI shorthand
0	Normal cartilage.
1	Signal heterogeneity/softening or swelling; surface intact.
2	Superficial fissuring/fraying/partial-thickness defect <50% cartilage depth.
3	Deep fissuring/partial-thickness defect >50% cartilage depth, not exposing bone.
4	Full-thickness cartilage loss with exposed subchondral bone, often edema/cysts.

Grade 0 - normal cartilage

> **F:** Articular cartilage of the [surface/compartment] is preserved in thickness and signal without focal fissuring, delamination, or subchondral marrow abnormality.

> **I:** No chondral defect of the [surface/compartment].

Grade 1 - chondral signal change / softening

> **F:** Mild focal cartilage signal heterogeneity/softening of the [surface/compartment] with preserved cartilage thickness and intact articular surface. No subchondral marrow edema.

> **I:** Grade 1 chondromalacia of the [surface/compartment].

Grade 2 - superficial partial-thickness chondral loss

> **F:** Superficial chondral fissuring/fraying of the [surface/compartment] involving less than 50% of cartilage thickness. No full-thickness defect or subchondral marrow edema.

> **I:** Grade 2 chondromalacia of the [surface/compartment].

Grade 3 - deep partial-thickness chondral loss

> **F:** Deep partial-thickness chondral fissuring/defect of the [surface/compartment] involving greater than 50% of cartilage thickness without exposed subchondral bone. [Mild adjacent subchondral marrow edema.]

> **I:** Grade 3 chondromalacia of the [surface/compartment].

Grade 4 - full-thickness chondral loss

> **F:** Full-thickness cartilage loss/fissuring of the [surface/compartment] measuring [X] x [Y] mm with exposed subchondral bone and [subchondral marrow edema/cystic change]. [Small unstable chondral flap/loose body if present.]

> **I:** Grade 4 chondromalacia/full-thickness chondral defect of the [surface/compartment].

Meniscus Tells

- Horizontal: degenerative, older patients; propagates along collagen fiber plane.
- Vertical longitudinal: traumatic, young patients; perpendicular to collagen fibers.
- Radial: functionally equivalent to meniscectomy at the tear site; disrupts circumferential hoop stress.
- Root tear: same biomechanical consequence as total meniscectomy; look for meniscal extrusion.
- Bucket handle: look for double PCL sign, absent bow-tie sign on consecutive sagittal images, and flipped fragment in the intercondylar notch.
- Ghost meniscus on sagittal = radial tear extending through the full width of the meniscal body.
- Meniscal extrusion >3 mm associated with root tears and accelerated cartilage degeneration.
- Meniscal flounce: wavy contour of the free edge on sagittal, normal variant when seen in a single plane.
- Discoid meniscus: >=3 consecutive sagittal bow-tie images (5 mm slice thickness) or meniscal body extends to the intercondylar notch on coronal.

Normal / Negative Templates

Normal menisci

> **F:** The medial and lateral menisci demonstrate normal morphology and low signal intensity without surface-reaching signal abnormality, displaced fragment, or meniscal extrusion. Meniscal roots are intact.

> **I:** Normal menisci bilaterally. No meniscal tear.

Normal ACL

F: The anterior cruciate ligament demonstrates normal fiber orientation, signal intensity, and tension from the tibial eminence to the lateral femoral condyle. No fiber discontinuity or surrounding edema.

I: Intact ACL.

Normal knee

F: The medial and lateral menisci are intact without tear or extrusion. The anterior and posterior cruciate ligaments demonstrate normal signal, orientation, and continuity. The MCL and LCL are intact. The extensor mechanism including the quadriceps tendon, patella, and patellar tendon is intact. Articular cartilage of all three compartments is preserved. No significant joint effusion. No marrow signal abnormality. No soft tissue mass.

I: Normal MRI of the knee.

Useful Caveats

- Meniscal signal: Grade 1/2 intrameniscal signal NOT reaching an articular surface = degenerative intrameniscal change, not a tear. Only surface-reaching signal on at least two consecutive images constitutes a tear.
- Post-meniscectomy: residual meniscus can show abnormal signal and irregular contour; compare to surgical report and prior imaging to differentiate expected postoperative change from retear.
- ACL mucoid degeneration: enlarged, intact ligament with diffuse increased T2 signal along its course; fibers are continuous; no clinical instability. Do not call a tear.
- Magic angle artifact: artifactual increased signal at 55 degrees to B0 in short TE sequences (PD, T1); verify on fluid-sensitive sequences before calling pathology.
- Transverse intermeniscal ligament: connects anterior horns; do not mistake for a free fragment or anterior meniscal tear on axial/sagittal images.

Meniscus

Horizontal tear

F: Horizontal fluid-signal tear of the [body/posterior horn] [medial/lateral] meniscus extending to the [superior/inferior] articular surface with mild meniscal degeneration. No displaced fragment.

I: Horizontal tear of the [medial/lateral] meniscus [body/posterior horn].

Vertical longitudinal tear

F: Vertical longitudinal fluid-signal tear of the [posterior horn/body] of the [medial/lateral] meniscus oriented perpendicular to the meniscal surface and extending [X] mm in craniocaudal dimension. The tear reaches the [superior/inferior/both] articular surface[s]. No displaced fragment or bucket-handle morphology.

I: Vertical longitudinal tear of the [medial/lateral] meniscus [posterior horn/body].

Radial tear / root equivalent

F: Radial tear of the [posterior horn/root] of the [medial/lateral] meniscus with truncation/ghosting on sagittal images and [X] mm peripheral extrusion of the meniscal body. Ghost meniscus sign is [present/absent] on sagittal imaging.

I: Radial/root tear of the [medial/lateral] meniscus with meniscal extrusion.

Complex tear

F: Complex multiplanar tear of the [posterior horn and body] of the [medial/lateral] meniscus with both horizontal and vertical components extending to [superior and inferior] articular surfaces. [Associated meniscal deformity/substance loss.] No displaced fragment.

I: Complex tear of the [medial/lateral] meniscus [posterior horn and body].

Bucket-handle tear

F: Displaced longitudinal tear of the [medial/lateral] meniscus with centrally displaced fragment lying in the intercondylar notch [deep to/adjacent to the PCL]. Donor meniscal body is diminutive with absent bow-tie sign on consecutive sagittal images. [Double PCL sign present on sagittal images.]

I: Bucket-handle tear of the [medial/lateral] meniscus with displaced fragment in the intercondylar notch.

Horizontal cleavage tear with parameniscal cyst

F: Horizontal cleavage tear of the [body/posterior horn] of the [medial/lateral] meniscus with fluid extending along the tear plane to the meniscal periphery. A [X] cm multiloculated T2-hyperintense/T1-hypointense cyst is present at the [medial/lateral] meniscocapsular junction, contiguous with the tear.

I: Horizontal cleavage tear of the [medial/lateral] meniscus with associated parameniscal cyst.

Meniscal flounce

F: Wavy/undulating contour of the free edge of the [medial/lateral] meniscus on sagittal images with otherwise normal meniscal signal and morphology. No surface-reaching signal abnormality.

I: Meniscal flounce of the [medial/lateral] meniscus, a normal variant. No meniscal tear.

Discoid meniscus with tear

F: The [lateral/medial] meniscus demonstrates discoid morphology with extension of the meniscal body to or near the intercondylar notch and >=3 consecutive sagittal bow-tie images. Superimposed [horizontal/complex] tear with fluid-signal extending to the [superior/inferior] articular surface. [Peripheral detachment/shift if present.]

I: Discoid [lateral/medial] meniscus with superimposed [horizontal/complex] tear.

Meniscal root tear

F: Tear of the [posterior/anterior] root attachment of the [medial/lateral] meniscus with fluid signal replacing the normal root fibers on [coronal/sagittal/axial] images. Ghost meniscus sign is present on sagittal imaging through the [posterior horn]. Meniscal body extrusion measures [X] mm beyond the tibial plateau margin. [Associated subchondral marrow edema/insufficiency fracture of the [medial/lateral] tibial plateau.]

I: [Medial/lateral] meniscal [posterior/anterior] root tear with [X] mm meniscal extrusion. Note: Biomechanically equivalent to total meniscectomy.

Meniscal extrusion

F: [Medial/lateral] meniscal body is extruded [X] mm beyond the tibial margin, associated with [root tear/advanced compartment chondrosis].

I: [Medial/lateral] meniscal extrusion, functionally significant at >=3 mm.

Cruciate Ligaments

ACL complete tear

F: Complete discontinuity of the ACL fibers with abnormal horizontal fiber orientation, intercondylar edema, and pivot-shift bone contusions of the lateral femoral condyle and posterolateral tibial plateau. [Anterior tibial translation/subluxation if present.]

I: Complete ACL tear with pivot-shift bone bruise pattern.

ACL sprain / partial tear

F: ACL is thickened and edematous with partial fiber disruption but preserved overall fiber continuity and orientation. No anterior tibial translation. [Surrounding intercondylar edema.]

I: ACL sprain/high-grade partial tear without complete rupture.

ACL mucoid degeneration

F: The ACL is diffusely enlarged with increased intrasubstance T2 signal and loss of normal low-signal fiber architecture. Fiber continuity is preserved with normal orientation from tibial to femoral attachment. No acute edema or pivot-shift bone contusion pattern. [Small intercondylar notch cyst may be present.]

I: ACL mucoid degeneration. Intact ligament without tear.

ACL ganglion cyst

F: Well-circumscribed T2-hyperintense/T1-hypointense lobulated cystic lesion within/adjacent to the ACL fibers measuring [X] cm. ACL fiber continuity is [preserved/disrupted]. No solid enhancing component.

I: ACL ganglion cyst. [ACL is intact / associated ACL tear].

PCL complete tear

F: Complete discontinuity of the posterior cruciate ligament with loss of normal arcuate morphology, fiber retraction, and surrounding hemorrhage/edema. [Associated posterior tibial translation.] [Concomitant posterolateral corner injury / multiligamentous pattern.]

I: Complete PCL tear. [Evaluate for multiligamentous knee injury.]

PCL sprain / partial tear

F: Focal thickening and increased T2 signal within the [proximal/mid/distal] PCL with partial fiber disruption. Overall ligament continuity and arcuate morphology are preserved. No posterior tibial translation.

I: PCL sprain/partial tear without complete rupture.

PCL mucoid degeneration

F: The PCL is diffusely thickened with increased intrasubstance T2 signal throughout its course. Fiber continuity is preserved with normal arcuate morphology. No acute edema or posterior tibial translation.

I: PCL mucoid degeneration without tear.

Collateral Ligaments / Posterolateral Corner

MCL sprain

F: Edema superficial and deep to the MCL with [mild thickening/partial fiber disruption] and preserved overall continuity. [Periligamentous fluid tracking.]

I: Grade [1/2] MCL sprain.

MCL grade 3 / complete tear

F: Complete disruption of the [superficial/deep] MCL at the [femoral/tibial/mid-substance] level with [X] mm fiber gap, surrounding hemorrhage, and periligamentous edema. [Concomitant deep MCL/meniscocapsular separation.] [Associated medial meniscus tear.]

I: Grade 3 (complete) MCL tear at the [femoral/tibial/mid-substance] level.

MCL bursitis

F: Fluid-signal collection deep to the superficial MCL and superficial to the deep MCL/medial joint line measuring [X] mm in thickness. No MCL fiber disruption or meniscal tear.

I: MCL bursitis. No ligament tear.

LCL sprain

F: Thickening and increased T2 signal within the lateral collateral ligament (fibular collateral ligament) with periligamentous edema. Fiber continuity is preserved from the lateral femoral epicondyle to the fibular head.

I: LCL sprain. No complete tear.

LCL complete tear

F: Complete disruption of the lateral collateral ligament at the [femoral/fibular/mid-substance] level with fiber retraction and surrounding hemorrhage/edema. [Associated posterolateral corner injury.]

I: Complete LCL tear. [Evaluate for posterolateral corner injury.]

Posterolateral corner injury

F: Injury to the posterolateral corner structures including [disruption/edema of the LCL], [thickening/tear of the popliteus tendon], [disruption of the popliteofibular ligament], and [edema/stripping of the arcuate ligament complex]. [Associated lateral meniscus tear.] [Associated cruciate ligament injury.]

I: Posterolateral corner injury involving the [LCL, popliteus tendon, popliteofibular ligament, arcuate complex]. Correlate for combined ligamentous instability.

Popliteus tendon tear

F: [Partial/complete] disruption of the popliteus tendon at the [musculotendinous junction/popliteal sulcus/femoral attachment] with surrounding edema and [tendon retraction if complete]. [Associated posterolateral corner injury/lateral meniscal tear.]

I: [Partial/complete] popliteus tendon tear. [Evaluate for posterolateral corner injury.]

Extensor Mechanism

Patellar tendon rupture

F: Full-thickness disruption of the [proximal/mid/distal] patellar tendon with [X] cm gap, surrounding hemorrhage/edema, and patella alta (Insall-Salvati ratio [X]).

I: Complete patellar tendon rupture with patella alta.

Patellar tendinopathy (jumper's knee)

F: Fusiform thickening and increased T2 signal within the proximal patellar tendon at the inferior patellar pole insertion without discrete fluid-signal tear. [Mild adjacent inferior patellar pole marrow edema.] Tendon continuity is preserved.

I: Patellar tendinopathy (jumper's knee). No discrete tear.

Quadriceps tendon partial tear

F: [Low-grade/high-grade] partial-thickness fluid-signal defect of the [superficial/deep] layer of the distal quadriceps tendon at its superior patellar insertion, involving approximately [X]% of tendon thickness. Remaining fibers are intact. No patella baja.

I: [Low-grade/high-grade] partial-thickness quadriceps tendon tear.

Quadriceps tendon complete tear

F: Full-thickness disruption of the quadriceps tendon [X] cm proximal to the superior patellar pole with [X] cm tendon retraction, surrounding hemorrhage/edema, and patella baja (Insall-Salvati ratio [X]). [Suprapatellar hematoma/effusion.]

I: Complete quadriceps tendon rupture with tendon retraction and patella baja.

MPFL tear

F: Disruption and edema of the medial patellofemoral ligament at the [patellar/femoral/mid-substance] level with [complete fiber discontinuity/partial tearing and thickening]. [Associated lateral patellar subluxation/tilt.]

I: [Partial/complete] MPFL tear, compatible with [recent/prior] lateral patellar dislocation.

Patellar dislocation pattern

F: Disruption of the medial patellofemoral ligament at the [patellar/femoral/mid-substance] level. Bone contusion/marrow edema of the inferomedial patella and anterolateral lateral femoral condyle reflecting the characteristic transient patellar dislocation pattern. [Osteochondral/chondral shear injury of the medial patellar facet and/or lateral trochlea.] [Loose body in the [suprapatellar pouch/lateral gutter].] [Joint effusion/hemarthrosis.]

I: Findings consistent with transient lateral patellar dislocation with MPFL tear, characteristic bone contusion pattern, and [osteochondral injury/loose body].

Patellar sleeve fracture

F: Avulsion of a thin osseous shell from the [inferior/superior] pole of the patella with fluid-signal cleft at the avulsion site and surrounding soft tissue edema. [Patella alta if inferior pole avulsion.] The patellar tendon attaches to the avulsed fragment. This pattern is characteristic of a pediatric injury.

I: Patellar sleeve avulsion fracture of the [inferior/superior] patellar pole. Pediatric pattern injury.

Cartilage / Arthritis / Loose Bodies

Patellofemoral chondrosis

F: [Grade 2/3/4] chondromalacia of the [lateral patellar facet/medial patellar facet/trochlea] with [subchondral marrow edema/cystic change]. TT-TG distance measures [X] mm. Patella is [centered/laterally tilted/subluxed].

I: [Mild/moderate/severe] patellofemoral chondrosis with [maltracking features if present].

Medial compartment chondrosis

F: [Grade 2/3/4] chondral [thinning/fissuring/full-thickness loss] of the medial femoral condyle weightbearing surface and/or medial tibial plateau measuring [X] x [Y] mm. [Subchondral marrow edema/cystic change.] [Marginal osteophytes.] [Associated medial meniscal tear/extrusion.]

I: [Mild/moderate/severe] medial compartment chondrosis [with associated medial meniscal pathology].

Lateral compartment chondrosis

F: [Grade 2/3/4] chondral [thinning/fissuring/full-thickness loss] of the lateral femoral condyle weightbearing surface and/or lateral tibial plateau measuring [X] x [Y] mm. [Subchondral marrow edema/cystic change.] [Marginal osteophytes.] [Associated lateral meniscal tear.]

I: [Mild/moderate/severe] lateral compartment chondrosis.

Osteochondral lesion

F: Focal osteochondral defect of the [medial femoral condyle/lateral femoral condyle/trochlea] measuring [X] x [Y] mm with fluid signal undermining the fragment and adjacent marrow edema.

I: Unstable osteochondral lesion of the [location].

Acute osteochondral defect / fracture — tibial plateau

F: Acute full-thickness osteochondral defect of the weight-bearing medial tibial plateau ([central/anterior/posterior] aspect) measuring [X] x [Y] mm (AP x transverse) with [Z] mm depth. Overlying cartilage is disrupted with an associated osteochondral fragment that is [in situ/partially detached/displaced], with [fluid signal undermining/no fluid undermining] the fragment base. There is acute subchondral marrow edema and a [nondisplaced subchondral fracture line] beneath the defect. Margins are sharp and non-sclerotic, without a subchondral sclerotic rim or cyst, consistent with an acute injury. [A displaced osteochondral fragment/loose body measuring [X] mm is identified in the (intercondylar notch/posterior recess/suprapatellar pouch).] Joint effusion is present [with a fluid-fluid level/lipohemarthrosis, indicating an intra-articular fracture]. [Associated internal derangement: (ACL tear/meniscal tear/collateral injury).]

I: Acute osteochondral defect of the weight-bearing medial tibial plateau, [X] x [Y] mm, with subchondral marrow edema [and an (in situ/displaced) osteochondral fragment]. [Unstable: fluid undermining the fragment / displaced fragment with intra-articular loose body.] [Lipohemarthrosis consistent with intra-articular fracture.]

Spontaneous osteonecrosis / subchondral insufficiency fracture (SONK/SIFK)

F: Focal subchondral low-T1/high-T2 signal abnormality of the [medial femoral condyle weightbearing surface/medial tibial plateau] measuring [X] mm with a low-signal subchondral fracture line and surrounding marrow edema disproportionate to the degree of cartilage loss. [Overlying chondral fissuring/flattening.] [Small joint effusion.]

I: Subchondral insufficiency fracture of the [medial femoral condyle/medial tibial plateau] (SONK/SIFK). Correlate with acute onset pain.

Osteochondritis dissecans (OCD)

F: Well-demarcated osteochondral lesion of the [medial femoral condyle (classic lateral aspect)/lateral femoral condyle] measuring [X] x [Y] mm. [Stability assessment: fluid signal undermining the fragment / T2-bright rim around fragment / overlying cartilage breach / in situ cystic change = unstable features.] [Fragment is in situ/partially detached/displaced.] [Overlying articular cartilage is intact/disrupted.]

I: Osteochondritis dissecans of the [location], [stable/unstable based on MRI criteria]. [Loose body if displaced.]

Tricompartmental OA

F: Tricompartmental osteoarthrosis with cartilage loss greatest in the [medial/lateral/patellofemoral] compartment, marginal osteophytes, subchondral marrow edema/cysts, and small effusion/synovitis.

I: Tricompartmental osteoarthrosis, greatest in the [compartment].

Bone / Marrow

Bone contusion - pivot shift pattern (ACL tear)

F: Bone marrow edema pattern (T1-hypointense/T2-hyperintense marrow signal) involving the posterolateral tibial plateau and the lateral femoral condyle sulcus terminalis, consistent with a pivot-shift mechanism. [Associated ACL tear.]

I: Pivot-shift bone contusion pattern. [Evaluate ACL.]

Bone contusion - dashboard injury pattern

F: Bone marrow edema pattern involving the anterior proximal tibia and the anterior distal femur, consistent with a dashboard/hyperflexion mechanism with posterior-directed force on the proximal tibia. [Evaluate PCL.]

I: Dashboard-pattern bone contusions. [Evaluate PCL.]

Bone contusion - clip/valgus pattern

F: Bone marrow edema pattern involving the lateral femoral condyle and lateral tibial plateau from impaction, with contrecoup edema of the medial femoral condyle/tibial plateau. [Associated MCL sprain/tear and possible ACL injury.]

I: Valgus/clip injury bone contusion pattern. [Evaluate MCL and ACL.]

Bone contusion - hyperextension pattern

F: Bone marrow edema pattern involving the anterior tibial plateau and anterior femoral condyles from direct impaction in hyperextension. [Associated ACL and/or PCL injury.]

I: Hyperextension bone contusion pattern. [Evaluate cruciate ligaments.]

Tibial plateau fracture

F: Fracture of the [lateral/medial/bicondylar] tibial plateau with [X] mm of articular surface depression and [X] mm lateral cortical widening/split component. [Associated meniscal tear/entrapment.] [Surrounding marrow edema and hemarthrosis.] [Fracture line extends to the [tibial spine/posterior cortex].]

I: [Schatzker type X] [lateral/medial/bicondylar] tibial plateau fracture with [X] mm articular depression.

Stress fracture

F: Linear low-T1/low-T2 signal line in the [proximal tibial metaphysis/distal femoral metaphysis] oriented perpendicular to the cortex with surrounding ill-defined marrow edema (T1-hypointense/T2-hyperintense). No discrete fracture displacement. [Periosteal edema/reaction.]

I: Stress fracture of the [proximal tibia/distal femur]. Correlate with activity history.

Transient bone marrow edema

F: Diffuse ill-defined marrow edema (T1-hypointense/T2-hyperintense) involving the [medial femoral condyle/lateral femoral condyle/proximal tibia] without discrete fracture line, osteochondral defect, or focal subchondral collapse. Articular cartilage is preserved. No meniscal or ligamentous injury.

I: Transient bone marrow edema of the [location]. Consider transient osteoporosis or early AVN in appropriate clinical setting.

Common Other Diagnoses

Baker cyst (intact)

F: Well-defined T2-hyperintense/T1-hypointense fluid collection between the medial gastrocnemius and semimembranosus tendons measuring [X] x [Y] cm, communicating with the joint through the posteromedial capsule. No internal debris or peripheral enhancement to suggest complication.

I: Baker cyst (popliteal cyst).

Baker cyst leak/rupture

F: Multiloculated popliteal cyst between the semimembranosus and medial gastrocnemius tendons with inferior fluid tracking along the calf fascial planes and surrounding soft tissue edema. [Cyst wall irregularity/discontinuity.]

I: Leaking/ruptured Baker cyst.

IT band friction syndrome

F: Edema deep to the iliotibial band over the lateral femoral epicondyle without IT band tear. [Thickening of the distal ITB.] [Small fluid collection between the ITB and lateral femoral condyle.]

I: Iliotibial band friction syndrome.

Pes anserine bursitis

F: Fluid and edema in the pes anserine bursa deep to the sartorius/gracilis/semitendinosus tendons along the anteromedial proximal tibia. No tendon tear.

I: Pes anserine bursitis.

Ganglion cyst

F: Well-circumscribed T2-hyperintense/T1-hypointense lobulated cystic lesion [adjacent to the proximal tibiofibular joint/along the meniscocapsular junction/in the intercondylar notch] measuring [X] cm. No solid component or internal enhancement. [Contiguous with adjacent meniscal tear/joint capsule.]

I: Ganglion cyst of the [location].

Pigmented villonodular synovitis (PVNS)

F: Diffuse/nodular synovial proliferation with heterogeneous signal and foci of T1/T2 hypointensity from hemosiderin deposition (blooming on gradient echo) throughout the suprapatellar pouch, posterior recesses, and periarticular soft tissues. Moderate joint effusion. [Osseous erosions/pressure defects without marginal osteophytes.]

I: Pigmented villonodular synovitis (PVNS) / diffuse-type giant cell tumor of the knee.

Synovial chondromatosis

F: Multiple intra-articular loose bodies of varying size within the suprapatellar pouch, posterior recesses, and intercondylar notch, many demonstrating peripheral calcification/ossification and central marrow-equivalent signal. Synovial proliferation. [Pressure erosions on articular surfaces without significant cartilage loss.]

I: Synovial chondromatosis of the knee.

Hoffa fat pad impingement

F: Edema and increased T2 signal within the infrapatellar (Hoffa) fat pad, predominantly in the superolateral aspect. [Focal mass-like fibrosis/scarring within the fat pad.] No discrete meniscal tear or patellar tendon pathology.

I: Hoffa fat pad edema/impingement syndrome.

Plica syndrome (medial plica)

F: Thickened, edematous medial plica measuring [X] mm in thickness extending from the suprapatellar pouch along the medial patellofemoral joint. [Adjacent chondral fissuring/erosion of the medial patellar facet and/or medial femoral condyle from plica impingement.]

I: Symptomatic medial plica with [associated chondral changes if present]. Correlate with anterior knee pain on clinical exam.

Search Pattern

- **1. Check history, indication, and priors.**
- **2. Assess adequacy, technique, and limitations.**
- **3. Check localizers** for incidental pathology.
- **4. Check for effusion/collection** on fluid-weighted sequences: joint effusion, synovial thickening/proliferation; loose bodies; cysts around joint (Baker's cyst) and associated meniscal cysts; bursitis.
- **5. Assess bone cortex and marrow:** cortical breaks/fractures, osteophytes, loose bodies, heterotopic ossification; marrow replacement.
- **6. Assess tendons:**
- Medial (pes anserinus: sartorius, gracilis, semitendinosus from anterior to posterior).
- Lateral (ITB, LCL, biceps femoris tendon, popliteus tendon, popliteofibular ligament).
- Anterior/extensor mechanism (quadriceps tendon, patellar tendon).
- Posterior (hamstrings, soleus/gastrocnemius, plantaris).
- **7. Assess ligaments:**
- Morphology and signal on at least two projections (coronals + sagittals, axials to troubleshoot).
- ACL (some intrinsic fluid signal and striated appearance can be normal).
- PCL (should be completely black).
- MCL (superficial and deep fibers, attached to meniscus).
- LCL (look for associated posterolateral corner injuries).
- Posterolateral corner structures (arcuate ligament, LCL, popliteus muscle, fabellofibular ligament, popliteofibular ligament, posterolateral capsule).
- **8. Assess menisci:**
- PD correlated with fluid-sensitive images; use coronals and sagittals, axials helpful for bucket-handle/flipped meniscal tears.
- Both medial and lateral; start at intermeniscal ligament attachment, assess anterior horn, body, posterior horns to root.
- Normal morphology and signal; thickness; position (double PCL sign = displaced/flipped/extruded).
- Does internal signal extend to articular surface on more than one sequence?
- **9. Assess cartilage:**
- PD and fluid-sensitive; patella (axials and sagittals); trochlea (sagittals).
- Medial compartment (sagittals and coronals); lateral compartment (sagittals and coronals).
- Subchondral marrow changes as clues.
- **10. Assess subcutaneous tissues and musculature:** T1 precontrast, fluid-sensitive; atrophy, edema, inflammation, mass; popliteal artery and neurovascular bundle (aneurysms, entrapment, occlusion, mass).
- **11. Double check** marrow signal, muscles, and subcutaneous tissues.
- **12. Proofread report.**

Infection Search Pattern

- **1. Marrow/bone changes:**
- Marrow replacement (darker than muscle/disc on T1; geographic pattern more predictive than reticular/periarticular).
- Marrow edema (isolated finding raises concern in diabetics/high-risk).
- Bone erosion (loss of dark T1/T2 cortical line).
- **2. Surrounding inflammatory changes:** fascia, musculature, subcutaneous fat edema/enhancement.
- **3. Susceptibility suggesting air:** in collections, soft tissues, along fascial planes, in bone.
- **4. Abnormal enhancement:**
- Rim-enhancing collection = abscess.
- Superficial lesion without enhancement = ulcer.

- Enhancement along fascial planes = infectious/inflammatory fasciitis.
- Non-enhancement of normal structures = ischemia.
- Other patterns of enhancement may not be specific.
- **5. Differentiating osteomyelitis from neuropathic joint:**
- Joint effusion with marrow replacement on both sides of joint.
- Fluid collections/abscesses (periprosthetic, subperiosteal, intraosseous/Brodie's abscess).
- Sinus tract connecting abnormal marrow signal to skin surface.
- Disappearance of subchondral cysts on sequential imaging.
- Progressive marrow/soft tissue changes greater than expected on sequential imaging.

Source Notes

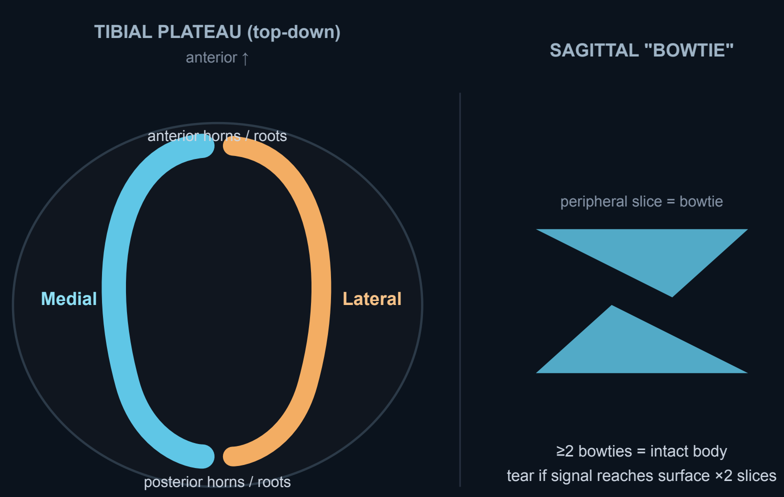
Built from the original knee MRI cheat-sheet plus review-level MSK MRI reporting patterns and the expanded shoulder cheat-sheet style. Includes bone contusion pattern recognition, posterolateral corner anatomy, meniscal tear classification, and patellar instability evaluation frameworks.

Test Yourself & Visual Pearls

Active-recall prompts and pattern recognition to sharpen your knee MRI eye.

Reading the meniscus

Orient yourself on sagittal sequences: the meniscus is a low-signal bowtie that you track from periphery to free edge, then confirm any abnormality on the orthogonal coronal plane.



Meniscus map. Medial and lateral menisci on the tibial plateau (anterior/posterior horns + roots). On sagittal, the body appears as a **bowtie** — ≥2 bowties = intact body; call a tear only when signal reaches a surface on ≥2 slices.

🔑 KEY POINTS **Meniscus rules**

- Two-slice-touch rule: a tear should be confirmed on ≥2 consecutive slices (or on both planes) before you call it — single-slice signal is usually a mimic or partial volume.
- Signal grading: grade 1 = globular intrasubstance signal (not surfacing); grade 2 = linear signal not reaching an articular surface; grade 3 = signal contacting an articular surface = a true tear.
- Bowtie counting: a normal meniscus shows ≥2 bowtie segments on contiguous 4–5 mm sagittal slices; fewer than 2 suggests a torn/displaced or post-meniscectomy meniscus.

- Extrusion: meniscal body displacement ≥3 mm beyond the tibial margin signals meniscal dysfunction (root tear, complex/radial tear, or degeneration).

? QUIZ **Why does grade 2 signal NOT count as a tear, and what changes it to grade 3?**

? QUIZ **What is the significance of meniscal extrusion ≥3 mm?**

Rapid review

Quick cases spanning menisci, ligaments, cartilage, and patellofemoral measurements.

? QUIZ **A radial tear of the meniscus produces which two classic signs, and why does it matter functionally?**

? QUIZ **How does a medial meniscus posterior root tear present, and what is its downstream consequence?**

? QUIZ **What two signs confirm a bucket-handle meniscal tear?**

? QUIZ **What is a ramp lesion and why is it easy to miss?**

⚠ PITFALL **Beware the normal structures that mimic meniscal tears.**

The meniscofemoral ligaments (Humphrey, anterior to PCL; Wrisberg, posterior to PCL), the transverse intermeniscal ligament (mimics an anterior-horn tear at its attachment), and the popliteomeniscal fascicles with the popliteus tendon (normal cleft at the lateral meniscus posterior horn) are all classic pseudo-tears. Follow each structure across slices.

? QUIZ **What is a meniscal flounce and is it pathologic?**

? QUIZ **Distinguish the primary from secondary MRI signs of an ACL tear.**

? QUIZ **Where are the pivot-shift bone contusions, and what bony avulsion clinches an ACL injury?**

? QUIZ **What does a thickened, "celery-stalk" ACL with intact fibers represent?**

? QUIZ **What is the arcuate sign and what injury does it flag?**

? QUIZ **How is an MCL injury graded on MRI?**

⚠ PITFALL **Don't let a subchondral fracture masquerade as osteonecrosis.**

"SONK" (spontaneous osteonecrosis of the knee) is now understood as a subchondral insufficiency fracture — a subchondral low-signal line with surrounding marrow edema, classically the medial femoral condyle, frequently downstream of a medial meniscus root tear. True avascular osteonecrosis has a different (epiphyseal, serpiginous) pattern.

? QUIZ **What defines an Outerbridge grade 4 cartilage lesion and an unstable osteochondral fragment?**

? QUIZ **After a lateral patellar dislocation, what bone-bruise pair and ligament should you check?**

? QUIZ **Which patellofemoral measurements predict instability, and their thresholds?**